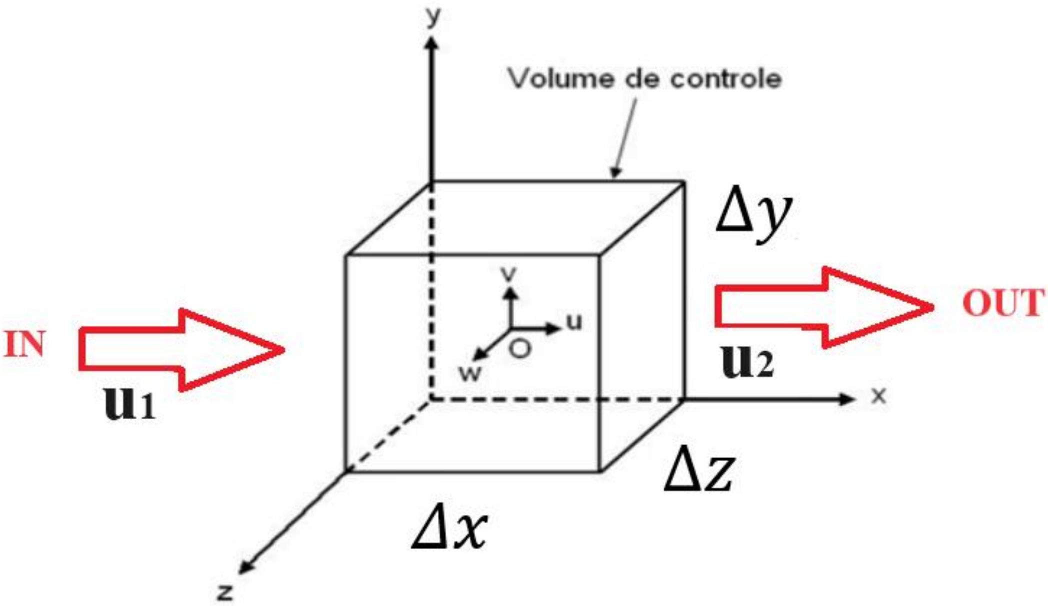
Leis de Conservação da Mecânica dos Fluidos aplicadas à Hidrodinâmica do Navio

I – Lei da Conservação de Massa

Vol . .

p τ - -

. . x



$p_{rxw} \quad p_{l} \quad p_{xer}$

-

$p \quad \tau x D$

$x \quad \frac{}{w} \quad D \quad C$ -

$p_{rxw} \quad \tau \; x \; D$

$p_{l} \quad \tau \; x \; D$

- -

$p_{xer} \quad \frac{p}{w} \quad C \frac{\tau}{w}$ - -

$p_{rxw} \quad p_{l} \quad p_{xer}$

$\tau \ x \ D \quad \tau \ x \ D \quad \frac{\tau}{w} \ - \ -$

$- \ -$

$D \quad D \quad -$

$\tau \ x \quad \tau \ x \quad - \quad \frac{\tau}{w} \ - \ -$

$\tau \ x \quad \tau \ x \quad \tau x$

$\tau x \quad - \quad \frac{\tau}{w} \ - \ -$

$\tau x \quad - \quad C \ \tau y \quad - \quad C \ \tau z \quad - \quad \frac{\tau}{w} \ - \ -$

$y \quad z$

$- \ -$

$\frac{\tau x}{} \quad C \frac{\tau y}{C} \quad C \frac{\tau z}{} \quad \frac{\tau}{w}$

$. \quad \frac{-}{-} \quad \frac{-}{-} C$

$$\frac{\tau x}{} \; C \frac{\tau y}{} \; \frac{\tau z}{} \qquad \frac{\tau}{w}$$

$$\tau \qquad \frac{\tau}{w}$$

$$\frac{}{} \; C \frac{}{} \; \frac{}{}$$

u, v e w *x, y e z*

$$\tau \quad - \quad g_D \qquad \frac{g}{gw} \qquad \tau C g \; _{ro}$$

$$\frac{\tau}{w}$$

$$\tau$$

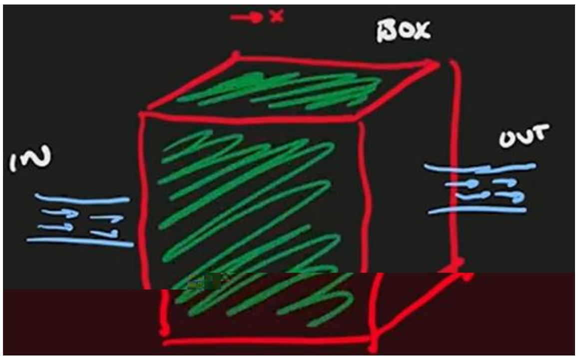
$$\frac{\tau x}{} \; C \frac{\tau y}{} \; \frac{\tau z}{} \quad \tau \; \frac{x}{} \; C \frac{y}{} \; \frac{z}{}$$

$$\frac{x}{} \; C \frac{y}{} \; \frac{z}{}$$

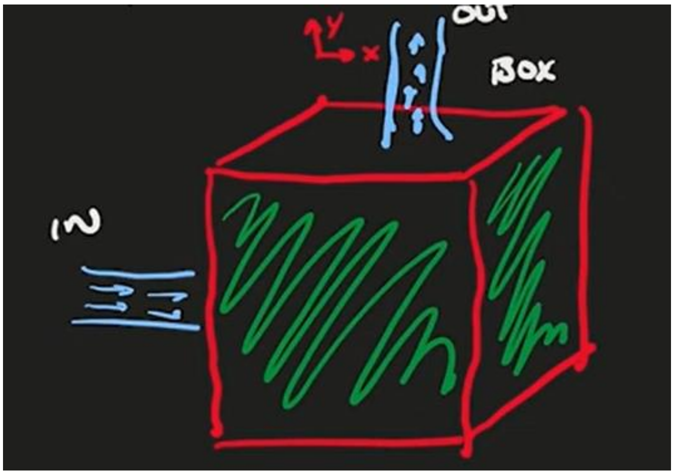
-

$\underline{x}$

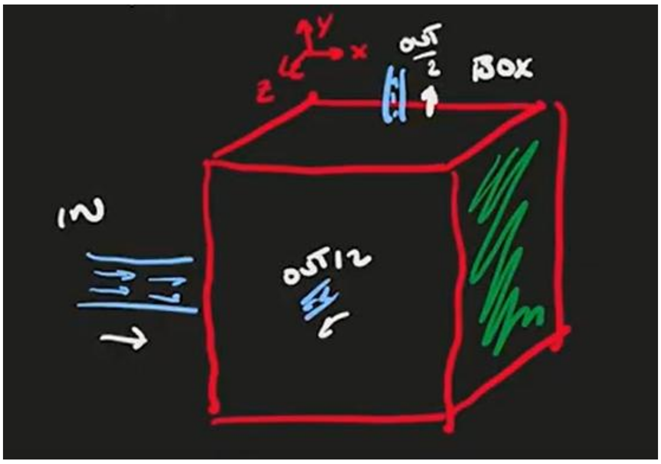
$\underline{y}$ $\underline{z}$



$\underline{x}$



$C\underline{x}$ $C\underline{y}$



$C\frac{x}{\quad}$ $C\frac{y}{\quad}$ $\frac{z}{\quad}$

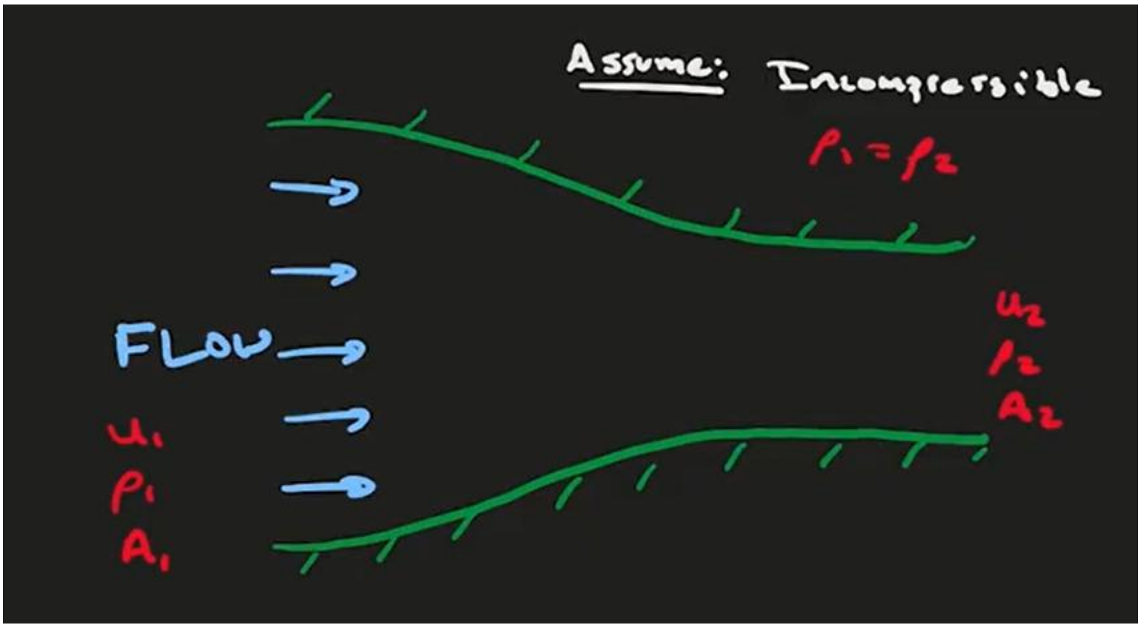
II – A lei de conservação da quantidade de movimento

$$\frac{p}{w} \quad C$$

A derivada material, derivada substantiva ou derivada total

$$\tau \frac{x}{w}$$

(u e u) τ *hG*τ *(A e A)*



$\tau \quad x \quad D \qquad \tau \quad x \quad D$

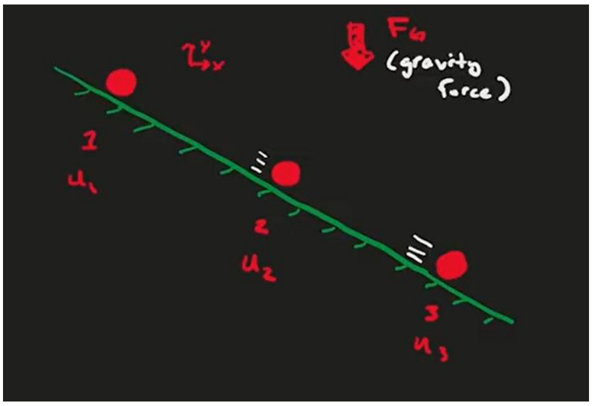
$\tau \qquad C\tau$

$x \quad D \qquad x \quad D$

$D \qquad D \qquad x \qquad x$

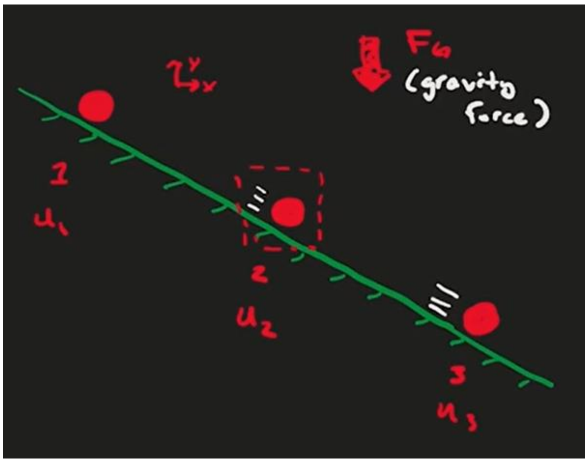
$\frac{x}{w}$

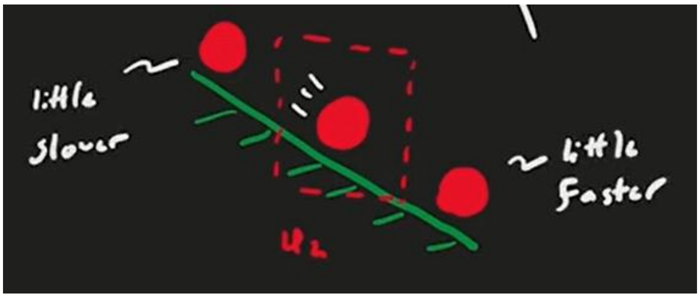
$\frac{x}{w}$



$x \qquad x \qquad x$


u
Δu
Δt
t
a = Δu / Δt


y
x
F_G
(gravity force)
1
u_1
2
u_2
3
u_3


~ little slower
~ little faster
u_2

$$\frac{x}{w} \quad \frac{x}{w}\frac{}{} \quad \frac{}{w}\frac{x}{} \quad x\frac{x}{}$$

$$\frac{x}{w} \quad \frac{x}{w}\frac{}{} \quad \frac{}{w}\frac{x}{} \quad y\frac{x}{}$$

C

$$d \quad \frac{x}{w} \quad x\frac{x}{} \quad y\frac{x}{} \quad z\frac{x}{}C$$

DvCdfhchud hvCqdvCgluh hvĆdChĈdCvhu rCuhvs hf lydp hq hC

$$d \quad \frac{}{w} \quad x \frac{y}{} \quad y \frac{y}{} \quad z \frac{y}{} C$$

C

$$d \quad \frac{z}{w} \quad x \frac{z}{} \quad y \frac{z}{} \quad z \frac{z}{} C$$

C

dvrCrChvfrdp hq rCvhmdCshup dqhq h.Chp vhC

$$C \frac{x}{w} \quad .C \frac{y}{w} \quad ChC \frac{z}{w}$$

Forças atuantes em um elemento fluido

$$\frac{\frac{p}{o}}{w} \quad C \frac{}{o}$$

$$\frac{p}{ro} \quad \tau$$

$$\frac{\tau}{w} \quad C \frac{}{o}$$

$$\tau \frac{}{w} \quad C \frac{}{o}$$